Department of Computer Science & Cybersecurity

Digital Forensics — Course Study Notes

Unit 2: Digital Evidence & Acquisition

Topic Focus: 2.4 Evidence Handling Procedures & 2.5 Data Acquisition Techniques (Live & Dead)

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1 Introduction

In digital forensics, the integrity of digital evidence is paramount. Any alteration, deletion, or contamination of data during the forensic lifecycle can render it inadmissible in a court of law. This unit focuses on two critical operational domains of forensic investigations:

1. **Evidence Handling Procedures (2.4):** The administrative and operational protocols followed from the moment evidence is identified until it is presented in court.
2. **Data Acquisition Techniques (2.5):** The technical methodologies utilized to capture and preserve digital assets from dynamic (Live) or static (Dead) systems.

2 2.4 Evidence Handling Procedures

The core objective of evidence handling is maintaining integrity, preventing contamination, and ensuring a verifiable **Chain of Custody**.

2.1 Identification and Collection

Before touching any system, an investigator must identify all potential sources of evidence and document the scene thoroughly.

- **Initial Assessment:** Determine the scope of the incident. Identify computers, mobile devices, servers, cloud storage, external media, and IoT devices.
- **Scene Documentation:** Take high-resolution photographs of the physical scene, system screens, cabling, and serial numbers. Document the state of the machine (On, Off, Suspended, or Locked).
- **Collection Order:** Collect physical items in a way that minimizes contamination. Use clean latex gloves to prevent transferring physical fingerprints or DNA to devices.

2.2 The Chain of Custody

The **Chain of Custody (CoC)** is a chronological, written record of custody, control, transfer, analysis, and disposition of physical or electronic evidence. If the defense can prove a gap in the CoC, the evidence is highly likely to be thrown out of court.

Every entry in the Chain of Custody must document:

1. **Who** took possession of the evidence.

2. **What** the evidence item is (description, model, serial number).
3. **When** the transfer occurred (precise date and time).
4. **Where** the evidence was secured or transported.
5. **Why** the transfer occurred (e.g., "Transport to lab for imaging").

Standard Chain of Custody Template Form

Case Number: 2026-DF-0941 **Investigator:** Det. Alex Chen
Item No: 01 **Description:** Samsung 1TB External SSD (S/N: MZ-77E1T0B)

Date/Time	Released By	Received By	Purpose / Action Taken	Location
16/07/2026 10:15	Scene Recovery	Det. Alex Chen	Collected at crime scene	Server Rm, Bldg 4
16/07/2026 11:45	Det. Alex Chen	Evid. Custodian	Logged into secure storage	Forensic Lab Vault
17/07/2026 09:00	Evid. Custodian	Analyst S. Kumar	Bit-stream imaging (E01)	Lab Workstation 3

2.3 Packaging, Transportation, and Storage

Physical and environmental hazards can easily corrupt digital components. Proper packaging and transportation are crucial.

- **Anti-Static Packaging:** Hard drives and circuit boards are highly sensitive to Electrostatic Discharge (ESD). They must be packed in anti-static Faraday bags.
- **Faraday Shielding:** For mobile phones, tablets, or wireless-enabled devices, packaging in a **Faraday Bag** is mandatory. This blocks all incoming/outgoing radio waves (cellular, Wi-Fi, Bluetooth), preventing remote-wipe commands or incoming data modification.
- **Transportation Safety:** Secure devices in padded, shock-absorbent containers. Avoid extreme temperature fluctuations during transport, as condensation can damage internal electronics.
- **Secure Storage:** Evidence must be stored in a climate-controlled, secure room or vault with restricted access. Every entry and exit to the storage facility must be logged.

3 2.5 Data Acquisition Techniques

Data acquisition is the process of retrieving digital information from physical storage media or volatile memory in a forensically sound manner. Acquisition is broadly categorized into **Live (Volatile)** and **Dead (Non-volatile)** acquisition.

3.1 Order of Volatility

When acquiring evidence, investigators must prioritize the collection of data based on how quickly it might be lost. This is defined by **RFC 3227 (Order of Volatility)**:

1. **Registers, Cache, and Active CPU States** (Changes in nanoseconds)
2. **Routing Tables, ARP Cache, Process Table, Kernel Statistics, RAM** (Changes in microseconds)
3. **Temporary File Systems / Swap Space** (Changes in minutes)
4. **Hard Disk Drives (HDD/SSD) and Storage Media** (Persists until overwritten)
5. **Remote Logging and Monitoring Data** (Persists on secondary servers)
6. **Physical Configuration and Network Topology** (Relatively static)

3.2 Live Acquisition (Volatile Data)

Live acquisition captures data from a system that is actively **powered on** and running.

3.2.1 Characteristics & Use Cases

- Captured when encrypted drives (BitLocker, VeraCrypt) are unlocked in memory.
- Essential for retrieving transient states, active network connections, logged-in sessions, and running malware processes.
- Used heavily in Incident Response (IR) where downtime is not acceptable for critical business servers.

3.2.2 Key Challenges

- **The Observer Effect:** Running an acquisition tool on a live system alters the system state (modifies RAM, updates file access times, and runs new processes).
- **Malware Interference:** Rootkits and sophisticated malware can detect forensic tools and actively lie, hide processes, or trigger self-destruct scripts.

3.2.3 Common Live Acquisition Tools

- **FTK Imager Lite:** A portable version of FTK Imager designed to run from an external USB drive to capture RAM and physical disks.
- **DumpIt / Belkasoft RAM Capturer:** Dedicated tools for dumping physical memory (RAM).

- **Volatility Framework:** An advanced command-line tool used to analyze the captured RAM dump offline.

3.3 Dead Acquisition (Static / Post-Mortem)

Dead acquisition takes place when the target machine is **powered off**. It is the most traditional and forensically secure method.

3.3.1 Characteristics & Process

- The target storage drive is physically removed from the machine (where possible).
- The drive is connected to a **Write Blocker** to ensure no write commands reach the evidence media.
- A bit-stream image (exact replica of every sector, including unallocated space, slack space, and deleted files) is created.

3.3.2 Write Blockers

Write blockers are critical to dead acquisition. They allow read commands but block write commands.

- **Hardware Write Blockers:** Physical bridging devices between the forensic workstation and the evidence drive (e.g., Tableau, WiebeTech). Highly reliable.
- **Software Write Blockers:** Operating system-level blocks (e.g., modifying the registry in Windows to block writes to USB devices). Economical but less secure than hardware solutions.

3.3.3 Forensic Image Formats

During a dead acquisition, the copy is stored in specific forensic image formats to guarantee data integrity:

- **Raw (DD/RAW):** A sector-by-sector clone with no metadata or compression. Extremely versatile but takes up a large amount of storage. File extensions are typically `.raw`, `.dd`, or `.img`.
- **Advanced Forensic Format (AFF):** An open-source extensible format that supports metadata, compression, and built-in MD5/SHA-1 hashing.
- **EnCase Evidence File (E01):** The industry standard developed by Guidance Software. It contains the acquisition metadata (case info, examiner name, acquisition date) and embeds cryptographic hashes at regular intervals to detect and report corruption instantly.

3.4 Live vs. Dead Acquisition: Comparative Matrix

Below is a structured comparison to help students understand when to deploy each technique.

Table 1: Comparative Matrix: Live vs. Dead Acquisition

Metric	Live Acquisition (Dynamic)	Dead Acquisition (Static)
System State	Machine is powered on and running	Machine is powered off
Main Focus	RAM, Network Connections, Active Processes	Non-volatile storage (HDD/SSD, USB)
Risk Level	High (Observer effect, malware tampering)	Negligible (Protected by Write Blockers)
Completeness	Captures active/hidden transient states	Captures deleted/unallocated sectors
Admissibility	Subject to challenge; must be meticulously logged	Readily accepted in courts of law
Common Tools	FTK Imager Lite, DumpIt, WinPmem	Hardware Write Blockers, EnCase, FTK

4 Student Review & Scenario Questions

Review Questions for Practice

1. **Question 1 (Concept Verification):** Explain why a forensic examiner should avoid booting a suspect's computer directly to look for evidence files. What alternative procedure should be followed?
2. **Question 2 (Chain of Custody):** Imagine you are an investigator. You recovered a USB flash drive from an office, imaged it on-site, and then shipped the physical drive to the headquarters lab. Outline the Chain of Custody entries that must occur during this timeline.
3. **Question 3 (Acquisition Choice):** You are dealing with an employee suspected of corporate espionage. The employee's desktop computer is currently on, and you notice a background process running. How would your acquisition approach differ if the drive was fully encrypted with BitLocker versus unencrypted?

Quick Reference Cheat-sheet for Students

- **Rule of Thumb 1:** Always document the scene *before* touching any hardware.
- **Rule of Thumb 2:** Preserve the volatile memory first (Order of Volatility!).
- **Rule of Thumb 3:** Never conduct forensic analysis directly on original evidence; always make a bit-stream copy and work on the duplicate.
- **Rule of Thumb 4:** Calculate and verify cryptographic hash values (MD5, SHA-256) of both the original media and the forensic copy to ensure 100% mathematical parity.